EXPLAINABLE AI FOR SHORT-TERM WIND-POWER FORECASTING

Evaluating SHAP Explanation Stability Across Wind Regimes and Temporal Periods

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1. Executive Summary

AI-based models can support short-term wind-power forecasting, but forecast accuracy alone does not establish whether their explanations are consistent across operating conditions. This proposal presents a quantitative, retrospective comparison of model performance and SHAP explanation stability for 10-minute-ahead turbine active-power forecasts. The primary models are Linear Regression as a simple baseline, XGBoost as a boosted-tree model, and LSTM as a sequence model. The core contribution is an auditable evaluation protocol rather than a new forecasting architecture or a claim that SHAP provides causal explanations.

The primary dataset is the public 10-minute SCADA record for the six-turbine Kelmarsh wind farm. The planned chronological design uses 2017–2022 for training, 2023 for validation, and 2024 as the final test period, subject to a data-integrity audit. In particular, the 2023 and 2024 v4 files will be checked against the reconstruction, signal-onset, and flag materials deposited in 2026; the original files will not be de-duplicated by keeping the last repeated timestamp [1]. The exact target and usable input channels will be confirmed from the official signal mapping before training.

Forecast performance will be reported using MAE, RMSE, and R^2 on common chronological test observations, overall and by wind-speed regime. SHAP explanations will be summarized globally and for selected individual predictions. Explanation stability will be assessed primarily by feature-rank agreement across chronological test windows within each regime, with top-feature overlap and attribution dispersion as secondary measures. A perturbation or grouped-ablation check will be used as a limited faithfulness test. The study will distinguish differences between wind regimes from instability within comparable conditions and will interpret conclusions only within the selected data, models, and horizon.

2. Research Problem

Wind-power output changes with meteorological conditions and turbine operating states. Short-term forecasts are useful for operational planning, yet the mapping from recent SCADA and weather measurements to future power is nonlinear and time-dependent. Models such as gradient-boosted trees and recurrent neural networks can learn patterns from multivariate records, but a low forecast error does not by itself show which recorded inputs contributed to a prediction or whether those attributions are consistent under different conditions.

Explainable-AI methods, including SHAP, have already been applied to wind-power forecasting, and research has proposed quantitative ways to evaluate the trustworthiness of explanations[2]. However, feature contributions can depend on the model, the reference/background data, and correlations among inputs. Wind-speed regimes and temporal periods can also differ in their data distributions. Consequently, a feature ranking from one test period may not be reproduced in another, even when aggregate forecast accuracy appears strong.

The research problem is therefore not a general absence of XAI in wind forecasting. It is the need for a focused, reproducible assessment of ;

a) how model-based SHAP attributions vary between predefined low-, medium-, and high-wind-speed regimes,
b) how repeatable those attributions are across chronological test windows within comparable conditions,
c) whether the models with lower forecast errors also have more stable explanations.

SHAP outputs will be interpreted as model attributions, not as evidence that a feature physically caused a change in power.

3. Research Purpose and Expected Contribution

The purpose of this study is to develop and evaluate a reproducible experimental protocol for 10-minute-ahead wind-power forecasting that assesses both predictive accuracy and the stability of SHAP explanations across wind regimes and chronological periods. The study will compare a simple Linear Regression baseline, XGBoost, and LSTM on a common, time-ordered SCADA dataset. It will not propose a new AI architecture or claim universal explainability.

The purpose has four connected parts:

- **Forecasting**: quantify how accurately each selected model predicts turbine active power one 10-minute interval ahead.
- **Explanation**: use SHAP to describe global feature contributions and contributions to selected individual test forecasts.
- **Stability and faithfulness**: compare explanations across wind regimes and chronological windows, and test model reliance on highly attributed feature groups through a limited perturbation or ablation analysis.
- **Integration**: examine whether predictive performance and explanation stability align across the evaluated models and conditions, without presuming that the most accurate model must have the most stable explanation.

The expected contribution is a transparent and reproducible assessment protocol, accompanied by empirical evidence for the selected dataset and models. The contribution is methodological and comparative: it is intended to show when explanations are consistent, when they change with conditions, and how those patterns relate to forecast errors.

4. Research Gap and Related Work

The literature establishes that XAI is already used in wind-power forecasting [2]. Compare several explanation techniques and propose quantitative measures for assessing explanation trustworthiness. Other studies have applied SHAP to forecasting models, including deep-learning approaches. A review of XAI in wind-energy systems identifies ongoing challenges that include limited temporal explainability, dataset limitations, and computational constraints [3]. A broader 2026 review of trustworthy renewable-energy forecasting also treats explanation assessment as one part of system-level trust rather than equating a feature-importance plot with trustworthiness [4].

Accordingly, this proposal does not claim that wind-energy XAI or explanation stability is entirely unexplored. The narrower gap is the need for a reproducible, time-aware experiment that measures SHAP attribution stability both across wind-speed regimes and across chronological test windows, while reporting forecasting error for the same models and observations. The experiment will also make dataset provenance and model-input availability explicit, because data defects or future-information leakage could otherwise undermine both forecast metrics and explanations.

Table 1: Prior Research and Study Gap

What prior work establishes	Remaining issue address here
XAI methods have been applied to wind-power forecasting, and quantitative measures for assessing explanation trustworthiness have been proposed.[2]	Evaluate explanation stability quantitatively rather than treating a SHAP plot alone as evidence of stable or reliable explanations.
Reviews identify limited temporal explainability and a shortage of suitable wind-specific benchmark datasets [3]	Compare attributions across predefined wind-speed regimes and chronological test periods.
A broader review describes fragmented trustworthy-AI approaches, model-centred explanations, and challenges associated with data quality and heterogeneous evidence [4]	Document the dataset audit and use a fixed, time-aware evaluation procedure with inputs available at forecast time.
SHAP has been used to explain wind-power forecasts at global and individual-prediction levels.[2]	Relate attribution stability to forecasting errors for the same models and test observations.

The novelty of this study lies not in introducing XAI or SHAP to wind-power forecasting, but in applying a focused, time-aware evaluation of explanation stability. Prior work has assessed XAI trustworthiness in wind-power forecasting [2], while reviews identify challenges involving temporal explainability and wind-specific benchmark data [3] and broader issues of fragmented system-level trust and data quality [4]. Building on this work, the proposed study will compare SHAP attributions across predefined wind-speed regimes and chronological test windows, while evaluating forecast accuracy on the same models and observations. It will also document data provenance and ensure that model inputs were

available at the forecast time. The intended contribution is a reproducible, dataset-specific assessment of how forecast accuracy relates to explanation stability—not a claim to be the first study of XAI or explanation stability in wind energy.

5. Research Questions

The research questions separate forecasting performance, feature attribution, regime dependence, temporal stability, and the relationship between accuracy and explanation stability:

- RQ1. How accurately can Linear Regression, XGBoost, and LSTM forecast turbine active power 10 minutes ahead using multivariate SCADA and available meteorological inputs?
- RQ2. Which input features receive the largest global and local SHAP attributions in the selected forecasting models?
- RQ3. How do SHAP feature attributions differ among predefined low-, medium-, and high-wind-speed regimes?
- RQ4. Within each wind-speed regime, how stable are SHAP feature rankings and attribution magnitudes across chronological test windows and repeated model fits?
- RQ5. How do forecast performance and explanation-stability measures relate across the selected models and evaluation conditions, and does the model with the best forecast accuracy also have the most stable explanations?

6. Research Objectives

Main objective: To evaluate short-term wind-power forecasting performance and quantify the stability of SHAP explanations across wind regimes and chronological test periods using a reproducible, data-audited experimental design.

Specific objectives:

1. Audit and prepare the selected multivariate SCADA dataset, documenting signal availability, data-quality decisions, temporal coverage, and forecast-time feature availability.
2. Develop and compare a consistent core set of forecasting models: Linear Regression, XGBoost, and LSTM, for 10-minute-ahead turbine active-power prediction.
3. Apply SHAP to generate global feature summaries and local explanations for held-out forecasts using documented model-appropriate explainers and reference data.
4. Measure explanation stability across low-, medium-, and high-wind-speed regimes and across chronological evaluation windows using pre-specified quantitative measures.
5. Evaluate forecasting errors and explanation behaviour jointly, including a cautious assessment of whether better forecast accuracy coincides with more stable explanations.

7. Proposed Title and Key Concepts

Proposed title:

"Explainable AI for Short-Term Wind-Power Forecasting: Evaluating SHAP Explanation Stability Across Wind Regimes and Temporal Periods"

The study uses the following concepts:

- **Short-term forecasting**: prediction of the next 10-minute turbine active-power observation from information available at the forecast origin.
- **Multivariate time series**: sequential SCADA and documented meteorological inputs, where each turbine's lag sequence remains separate and ordered.
- **Forecasting models**: Linear Regression as a baseline, XGBoost for boosted-tree learning, and LSTM for sequential modelling.
- **SHAP attribution**: a model-based decomposition of prediction contributions, used for global and local interpretation; it is not a causal method.
- **Condition dependence**: differences in attribution patterns between wind-speed regimes.
- **Explanation stability**: repeatability of feature rankings and attribution magnitudes across chronological windows, repeated fits, or comparable samples within a regime.
- **Faithfulness check**: a limited perturbation/ablation test of whether features with high attribution are associated with greater changes in the model output; this does not establish physical causality.

7.1. Conceptual Framework

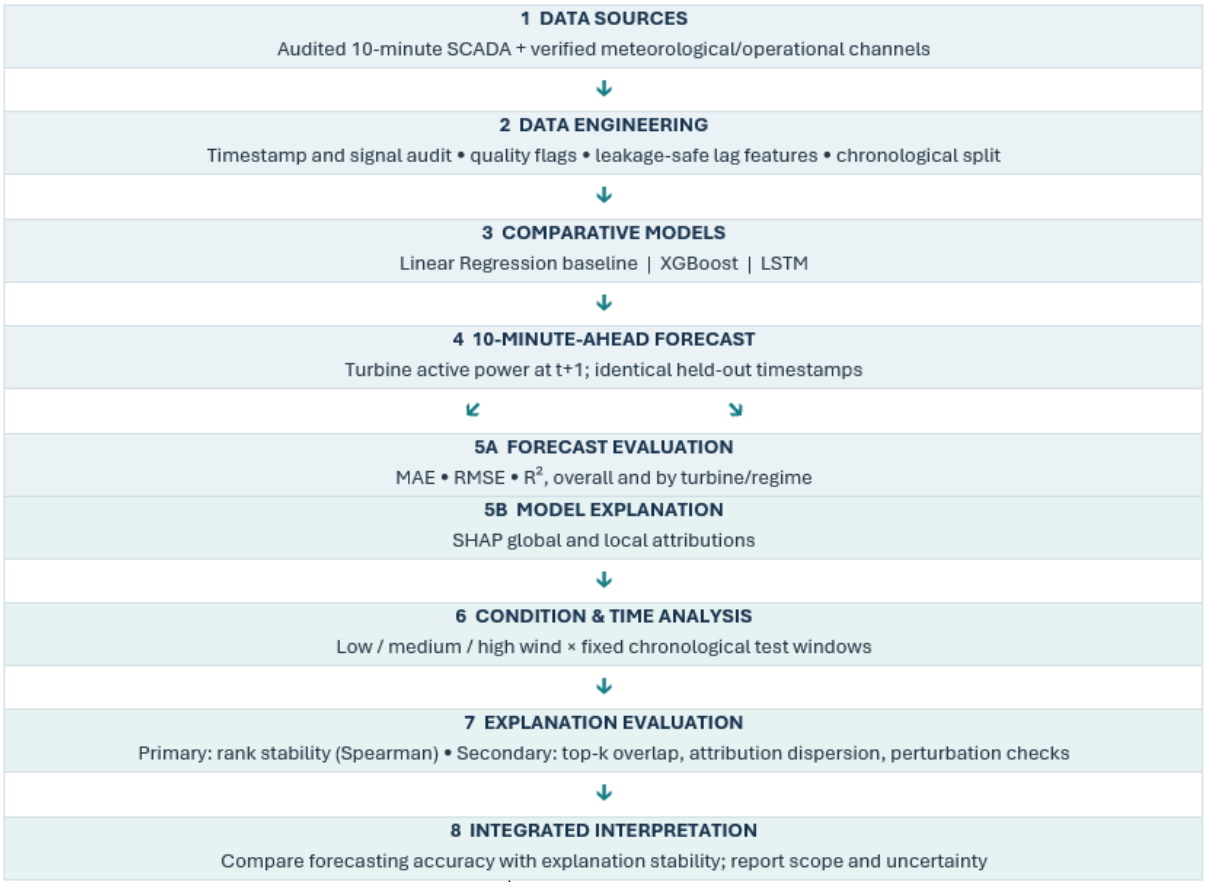


Figure 1:Proposed end-to-end analysis framework.

8. Data Collection Methods and Datasets

The study will use secondary, publicly available SCADA and meteorological/operational records; no new measurements will be collected from turbines. The primary source is the Kelmarsh wind-farm dataset released by Plumley and Takeuchi (2025a)[6],[7]. It contains 10-minute SCADA and event records for six Senvion MM92 turbines, grouped by year from 2016 to 2024, together with signal mappings and turbine metadata. The release is under CC BY 4.0. Not all signals are available for the full period, so the final input set will be selected only after inspecting the files and signal-onset information.

The proposed primary split is 2017–2022 for training, 2023 for validation, and 2024 for the final chronological test. The 2023 and 2024 files will use the audited reconstruction. The 2016 files may be used for schema familiarization but are not required for the primary experiment. All dataset versions, file hashes, signal mappings, filters, and transformations will be logged so that the analysis can be reproduced.

The target will be turbine-level active power at time t+1, representing a 10-minute-ahead forecast on the 10-minute data grid. Candidate inputs include lagged active power, wind speed and direction, and available turbine-operational channels such as rotor speed, generator speed, pitch angle, or nacelle direction. The exact signal names, units, valid ranges, coverage, and status meanings will be verified from the official documentation. Features will be restricted to information available by the forecast issue time; future observed SCADA values will not be used. ERA5 is reanalysis, not an operational forecast, so it will not be used as a future-weather predictor. Any future meteorological input would require a forecast product demonstrably available at issue time.

8.1. External Validation and Dataset Scope

Penmanshiel is proposed as a secondary, conditional external site test because its public record provides 10-minute SCADA and event data for a second UK farm. It will be used only if a documented common feature set and adequate time coverage can be established. The test procedure will be specified as transfer without retraining or fine-tuning; otherwise, it will be described as a separate site-specific experiment rather than external validation. Because the sites are in the same country and have related turbine technology, this comparison would assess site transfer, not global or cross-climate generalization. SDWPF is excluded from the core design to avoid expanding scope and because its precise site location and file-licence details require further verification.

8.2. Data Quality and Governance

1. Preserve raw downloads and record repository version, date accessed, licence, file checksum, and processing code.
2. Check timestamps, duplicate rows, missingness, signal onset, units, sentinel values, physical-range flags, and turbine event/status logs.
3. Do not automatically clip all negative or extreme values; review against the signal definition and event context, and document any exclusion or correction rule.

4. Fit imputation, scaling, and feature-selection transformations on training data only; apply those fitted transformations unchanged to validation and test data.
5. Use the intersection of consistently available features across selected turbines and periods for the main comparison; report any narrower per-signal analyses separately.
6. Attribute the public dataset under its stated CC BY 4.0 licence and do not redistribute modified source data without complying with the repository terms.

9. Research Approach and Design

The research adopts a quantitative, empirical, retrospective, and comparative design. It is a computational experiment in which forecasting models are trained and assessed on historical observations, then their prediction attributions are compared under pre-specified operating regimes and time windows. The design does not involve intervention in turbine operation and does not support causal claims about physical drivers.

Table 2:Research Design Summary

Design Element	Specification
Paradigm and approach	Quantitative; empirical analysis of secondary time-series data.
Primary unit	One turbine-time observation; sequence construction is performed separately within each turbine.
Target	Turbine active power at t+1 (10 minutes ahead), subject to signal and unit verification.
Core models	Linear Regression baseline; XGBoost; LSTM.
Primary validation	Chronological train 2017–2022 / validation 2023 / final test 2024; reconstructed and audited 2023–2024 source files.
Secondary validation	Penmanshiel only if features can be harmonized; report as site-transfer evidence, not broad generalization.
Main outputs	Forecast errors; global/local SHAP; within-regime temporal stability; perturbation-based faithfulness check.

9.1. Feature and Model Design

For Linear Regression and XGBoost, inputs will be represented as a fixed vector of lagged and currently available features. For LSTM, the same high-level variables will be provided as ordered sequences; sequences will never cross turbine boundaries or data gaps. A primary lookback length will be selected using training/validation data only and documented before the final test. Input transformations and tuning will use validation data, not the final test year.

Wind direction will be represented in a circularly appropriate form where used. Wind regimes will be defined before final test evaluation using turbine-specific power-curve or operating information where available. If reliable engineering cut-offs are unavailable, cut-offs will be derived from the training-period wind-speed distribution and labelled as statistical low-, medium-, and high-wind bands rather than universal physical operating states. The selected cut-offs will be frozen before testing, and sample counts by regime will be reported.

9.2. SHAP Configuration

SHAP will be applied with explainers compatible with each model architecture. The exact explainer, software version, background/reference sample, and feature representation will be recorded. The same high-level predictors will be used across models where possible,

and LSTM attributions across time steps will be aggregated to feature-level summaries using a pre-specified rule. Global explanations will be computed on a stratified, pre-specified subset of held-out observations if full-data computation is impractical; local explanations will be shown for representative test forecasts chosen by an explicit sampling rule. A sensitivity check will examine whether conclusions change under a reasonable alternative background sample.

10. Analysis and Interpretation Plan

10.1. Exploratory Data Analysis

EDA will describe data quality and the operating range before model fitting. It will include wind-speed and active-power distributions, the wind-speed–power relationship, correlations among candidate features, timestamp coverage, missingness, flagged observations, and temporal patterns. Analyses used to select variables, establish regime boundaries, or tune preprocessing will rely on development data; the final test year will remain reserved for evaluation.

10.2. Forecast Performance

All models will be scored on the same held-out turbine-time observations. MAE, RMSE, and R^2 will be reported overall, by turbine, and by wind regime where sample counts allow. MAE and RMSE will be expressed in the target's verified power unit. Training time may be reported as a secondary computational measure. MAPE will not be a primary metric because zero or near-zero turbine output can make percentage errors undefined or misleading [5].

Table 3:10-Minute-Ahead Model Performance

Model	Horizon	MAE	RMSE	R2	Training Time
Linear Regression	10 min	To be measured	To be measured	To be measured	To be measured
XGBoost	10 min	To be measured	To be measured	To be measured	To be measured
LSTM	10 min	To be measured	To be measured	To be measured	To be measured

A model will be described as more accurate only with reference to the evaluated horizon, split, and metric. For example, a lower RMSE means lower squared-error-weighted forecast loss on those observations; it does not alone prove that a model captured temporal dynamics better. Forecast comparisons will use paired time-aligned errors and uncertainty estimates that account for temporal dependence, such as block bootstrap intervals.

10.3. Global and Local SHAP Analysis

Global feature summaries will use the distribution and mean absolute SHAP magnitude on held-out observations, with signed contributions examined to understand whether feature values tend to push predictions upward or downward. Local analyses will explain a pre-selected sample of individual forecasts, including examples from each wind regime and from high-error cases. Attributions will be described as model contributions conditional on

the fitted model and reference distribution; correlated inputs may share attribution, and the results will not be interpreted as causal effects.

10.4. Stability and Faithfulness Analysis

The analysis will distinguish between cross-regime condition dependence and within-regime temporal stability. For each regime, the held-out period will be divided into pre-specified chronological windows (quarterly windows where sample counts permit). Global SHAP rankings will be compared across windows within the same regime. The primary stability statistic will be pairwise Spearman rank correlation; secondary statistics will be top-five Jaccard overlap and dispersion of feature-level SHAP magnitudes. Results will be reported as continuous estimates with uncertainty rather than forced into an unsupported universal stable/unstable cut-off.

- **Primary stability**: Spearman correlation of global feature rankings between chronological windows within each wind regime.
- **Secondary stability**: Jaccard overlap of top-five features and normalized dispersion of feature-level attribution magnitudes.
- **Seed sensitivity**: repeat stochastic LSTM fits under a pre-specified set of seeds, where computationally feasible, and separate seed variation from temporal variation.
- **Faithfulness check**: grouped perturbation or feature-ablation analysis to test whether removing/perturbing highly attributed feature groups changes model outputs or errors more than comparable low-attribution groups; interpret cautiously where features are correlated.
- **Regime contrasts**: describe how feature rankings and signed contributions differ between regimes without labelling every between-regime difference as instability.

10.5. Statistical Comparison and RQ5

Model errors and stability statistics will be compared on matched test windows and turbines. Resampling will preserve time blocks and, where appropriate, turbine grouping. The relationship between accuracy and stability will be summarized across models, regimes, and windows. Because the core model set contains only three model families, any model-level correlation will be treated as exploratory; it will not be presented as a general law that accuracy predicts explanation quality. The conclusion will state whether the best-accuracy model also ranks highest on the pre-specified stability measures in this experiment.

10.6. Interpretation Matrix

Table 4:Accuracy–Stability Interpretation Matrix

Forecast Accuracy	Explanation Stability	Interepretation within this study
Higher	Higher	Accurate candidate with comparatively repeatable attributions under the tested conditions.
Higher	Lower	Accurate forecasts, but attributions vary; explanatory claims require caution.
Lower	Higher	Repeatable attributions for a weaker forecasting model; stability does not compensate for poor accuracy.
Lower	Lower	Neither forecast skill nor explanation repeatability is satisfactory under the tested setup.

The matrix supports a balanced conclusion. A stable SHAP pattern is not automatically a reliable or physically true explanation; reliability claims will be limited to the operational stability and perturbation checks actually completed.

11. Limitations and Delimitations

The study is deliberately bounded to make the central experiment feasible. Its conclusions will apply to the selected data, models, input signals, wind-regime definitions, SHAP settings, and 10-minute horizon. The following limitations will be acknowledged:

- **Public-data dependence**: coverage, missingness, signal definitions, and recording quality are controlled by data providers. Data corrections and exclusions may affect findings; the provenance audit and public flag files reduce but do not eliminate this risk.
- **Single-site focus**: the core study uses Kelmarsh. Penmanshiel, if feasible, tests transfer to a second UK site but does not establish global or cross-climate generalization.
- **Incomplete physical observability**: SCADA may omit turbulence, wake effects, local terrain, curtailment, maintenance state, or other drivers. The models cannot explain information absent from the inputs.
- **SHAP and correlated inputs**: explanations depend on model, feature representation, reference data, and attribution assumptions. They are model-based, not causal ground truth.
- **Stability operationalization**: rank correlation, overlap, and attribution dispersion capture different properties; the selected measures and windows may affect the conclusion. No universal threshold is claimed.

- **Temporal non-stationarity**: chronological and rolling evaluation reveal some distribution changes but cannot guarantee future performance or eliminate all shifts between years.
- **Horizon and computation**: the core claim is limited to 10 minutes ahead. LSTM and repeated SHAP calculations may constrain the number of model configurations and robustness analyses.
- **Accuracy–explanation trade-off:** the best forecast model may not have the most stable explanations. The study reports this empirical comparison but does not claim to solve the general trade-off.

11.1. Delimitations

The study will not develop a novel neural architecture, make causal claims about meteorological drivers, conduct a human-subject trust study, or make claims about day-ahead and long-term forecasting. Random Forest and SDWPF are excluded from the core experiment to maintain alignment and feasibility; they may be considered only in future extensions.

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